

Digital Signal Analysis and Processing Exam Notes

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

1. Discrete-Time Signals and Systems

8 hrs · the largest chapter, **105 questions across the 45 papers** · worth about **12 of 80 marks** every year · **576 marks, 15.7% of the archive** · 10 syllabus sub-topics · numericals in the Ch1 companion.

Read this first:

- **Convolution is the single most-asked question in the whole subject:** some form of “find $y(n)$ given $h(n)$ and $x(n)$ ” appears in **34 of 45 papers** **TOP 34/45**, usually for 5 or 6 marks. If you learn one thing in this chapter, learn that.
- The chapter’s other four regulars, in order:
 - energy Vs power signal + is it periodic **TOP 10/45**,
 - test a given system for linearity / time-invariance / causality / stability **HOT 7/45**,
 - find the fundamental period of a given sequence **HOT 7/45**,
 - sampling an analog signal and writing down $x[n]$ **HOT 7/45**.
- Counted by *topic* rather than by exact question, system properties come up in **16** papers and sampling in **10**, so nothing here is optional.
- **The marks sit in the method, not the answer.** Almost every question in this chapter is a 3-line recipe applied to new numbers. Sections 1.3, 1.4, 1.8 and 1.9 each end with the recipe written out as steps: memorise those, not the worked examples.
- Tier chips count papers out of 45: **TOP n/45** ≥ 10 papers, **HOT n/45** 4–9, **PIN n/45** 1–3. **Bold** year codes are Regular exams, plain are Back; typewriter codes are EX 753 (BEX) papers.

1.1 What a discrete-time signal is, and how it is classified

Define and compare continuous-time and discrete-time signals; classify a given signal **2 3**

PIN 3/45 69 Bh, 73 Ch, 76 Ch · never asked alone, always as the 2-mark opening of a bigger question

By the independent variable

- **Continuous-time (analog):** $x(t)$ defined for a continuum of t .
 - Eg speech voltage at a microphone, video signal.
- **Discrete-time:** $x[n]$ defined only at discrete instants, n integer.
 - Instants need not be equally spaced, but in practice $t = nT$ is used.
 - Eg monthly sales of a corporation, daily stock-market average.
 - **$x[3.12]$ does not exist.** A DT signal is a *sequence*, not a function sampled finely.

By the value the signal takes

- **Continuous-valued:** amplitude takes any value in a range.
- **Discrete-valued:** amplitude comes from a finite set.
- **Digital signal** = discrete in *time* + discrete in *value*.
 - This is the only kind a processor can store, so A/D does both steps: sampling (Ch 1.10) then quantization (Ch 8).
- **Exam trap:** “discrete-time” and “digital” are not synonyms. Sampling alone gives a discrete-time, continuous-valued signal.

Define even and odd discrete-time signals; find the even and odd part of a given signal 2 3 4+5

HOT 4/45 71 Ch, 70 Asa, 71 Shr, 67 Mng · plus 76 Ch which asks the definition then a plot · worked instances in Ch1-num §7

Definitions

- **Even (symmetric):** $x[-n] = x[n]$.
- **Odd (antisymmetric):** $x[-n] = -x[n]$.
- An odd signal must have $x[0] = 0$, since $x[0] = -x[0]$.
- Any signal splits into exactly one even + one odd part.

The two formulas

$$x_e[n] = \frac{1}{2}(x[n] + x[-n]), \quad x_o[n] = \frac{1}{2}(x[n] - x[-n])$$

Derivation, if the question says “show that”

- Start from $x[n] = x_e[n] + x_o[n]$ (1)
- Put $n \rightarrow -n$: $x[-n] = x_e[-n] + x_o[-n]$ (2)
- Use $x_e[-n] = x_e[n]$ and $x_o[-n] = -x_o[n]$ in (2): (3)
- $x[-n] = x_e[n] - x_o[n]$
- (1)+(3) $\rightarrow x[n] + x[-n] = 2x_e[n]$.
- (1)-(3) $\rightarrow x[n] - x[-n] = 2x_o[n]$.

Method, every time

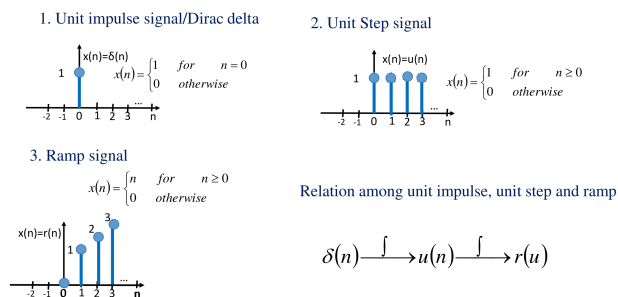
1. Write $x[n]$ as a sequence with the $n = 0$ sample marked.
2. Fold it to get $x[-n]$: same samples, index sign flipped.
3. Half the sum, half the difference, aligning on n .
4. **Check:** x_e must be symmetric about $n = 0$ and $x_o[0] = 0$.

1.2 Elementary discrete-time signals

Define and plot the unit step / unit impulse; explain their relation 1+2 3

PIN 3/45 73 Ch, 74 Ch, 69 Bh · the plotting half is done in Ch1-num §8

Elementary discrete-time signals



The three building blocks and the chain that links them. BB Sir, deck 1.7

The five standard sequences

- **Unit impulse** $\delta[n] = 1$ at $n = 0$, else 0.
- **Unit step** $u[n] = 1$ for $n \geq 0$, else 0.
- **Unit ramp** $r[n] = n$ for $n \geq 0$, else 0.
- **Real exponential** $x[n] = \alpha^n$: decays for $|\alpha| < 1$, grows for $|\alpha| > 1$, alternates sign when $\alpha < 0$.
- **Complex exponential** $x[n] = r^n e^{j\theta n} = r^n (\cos \theta n + j \sin \theta n)$.

Relations worth quoting (this is the marks)

$$\delta[n] = u[n] - u[n-1], \quad u[n] = \sum_{k=-\infty}^n \delta[k]$$

- Step is the running sum of the impulse; impulse is the first difference of the step.
- Ramp is the running sum of the step: $\delta \rightarrow u \rightarrow r$.
- **Sifting:** $x[n]\delta[n-k] = x[k]\delta[n-k]$, so $x[n] = \sum_k x[k]\delta[n-k]$. **This one line is what makes convolution work (§1.9).**

1.3 Energy and power signals

Define / compare energy and power signals, and classify the given signal 2+2 3 5

TOP 10/45 79 Bh, 69 Ch, 81 Bh, 74 Ash, 79 Ba, 82 Bh, 72 Ka, 68 Bh, 75 Ch, 74 Ma · in 81 Bh only the periodicity half is asked · always paired with §1.4 · worked in Ch1-num §5

Over a finite interval $n_1 \leq n \leq n_2$

$$E = \sum_{n=n_1}^{n_2} |x[n]|^2, \quad P_{av} = \frac{1}{n_2 - n_1 + 1} \sum_{n=n_1}^{n_2} |x[n]|^2$$

Over all time

$$E = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2, \quad P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

- **Energy signal:** $0 < E < \infty$, and then $P = 0$.
- **Power signal:** $0 < P < \infty$, and then $E = \infty$.
- **A signal can be neither** (Eg $x[n] = n$), and **never both**.

How to decide in one line

1. Finite-length or decaying sequence $\rightarrow$ energy signal.
2. Never-ending, bounded, non-decaying $\rightarrow$ power signal.
3. Every periodic signal is a power signal, with $P = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2$ over one period.
4. Every complex exponential $e^{j(\omega_0 n + \phi)}$ has $|x[n]| = 1$, so $P = 1$ and $E = \infty$: **power signal, no calculation needed**. That is the whole answer to 79 Ba and 82 Bh.

Standard results to quote

- $x[n] = \delta[n]$: $E = 1, P = 0 \rightarrow$ energy.
- $x[n] = u[n]$: $E = \infty, P = \frac{1}{2} \rightarrow$ power. **$P = \frac{1}{2}$, not 1:** the sum runs over $2N+1$ samples but only $N+1$ are non-zero.
- $x[n] = A$ (constant): $E = \infty, P = A^2 \rightarrow$ power.
- $x[n] = a^n u[n], |a| < 1$: $E = \frac{1}{1-a^2} \rightarrow$ energy.

Units

- Follow the signal: a voltage signal gives energy in V^2s and power in V^2 .

Sums you will need in every one of these (learn this block)

$$\sum_{n=-N}^N 1 = 2N+1, \quad \sum_{n=0}^N 1 = N+1$$

$$\sum_{n=0}^{N-1} a^n = \begin{cases} N, & a = 1 \\ \frac{1-a^N}{1-a}, & a \neq 1 \end{cases}$$

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a} \quad (|a| < 1), \quad \sum_{n=k}^{\infty} a^n = \frac{a^k}{1-a}$$

$$\cos \theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta}), \quad \sin \theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$$

1.4 Periodicity of a discrete-time signal

Determine whether the given sequence is periodic; if it is, find the fundamental period 2+2 3 4 5

HOT 7/45 80 Bh, 80 Ba, 78 Bh, 70 Ch, 69 Bh, 67 Mng, 66 Ma · another 10 papers ask it as the second half of the energy/power question · worked in Ch1-num §6

Definition

- $x[n]$ is periodic with period $N > 0$ if $x[n] = x[n + N]$ for all n .
- Smallest such N is the **fundamental period**. N must be an **integer**: that is the whole difference from the continuous-time case.

The rule, and where it comes from

- For $x[n] = e^{j\omega_0 n}$ to repeat: $e^{j\omega_0(n+N)} = e^{j\omega_0 n} \rightarrow e^{j\omega_0 N} = 1$
 $\rightarrow \omega_0 N = 2\pi m$.
- So $\frac{\omega_0}{2\pi} = \frac{m}{N}$ must be a **rational number**.
- **A discrete-time sinusoid is periodic only if $\omega_0/2\pi$ is rational.**
- Then $N = \frac{2\pi m}{\omega_0}$ with m the smallest integer making N an integer.

Method, every time

1. Read off ω_0 (the coefficient of n inside cos, sin or the exponent).
2. Form $\omega_0/2\pi$. Irrational $\rightarrow$ **aperiodic**, stop.
3. Rational = m/N in lowest terms $\rightarrow$ fundamental period is N .
4. Sum of two periodic signals: $N = \text{LCM}(N_1, N_2)$, and it is periodic only if N_1/N_2 is rational.

The three traps the papers actually set

- $x[n] = \sin(0.8n)$ or $e^{j(7/5)n}$: ω_0 has **no π in it**, so $\omega_0/2\pi$ is irrational $\rightarrow$ **aperiodic (66 Ma, 69 Bh)**.
- $x[n] = \cos(\pi n^2/8)$: n^2 , not $n \rightarrow$ not a sinusoid, test $x[n + N] = x[n]$ directly. It is periodic, $N = 8$ (**70 Ch**).
- $\sin(\pi n/\sqrt{2} + \pi/8)$: the $\sqrt{2}$ makes $\omega_0/2\pi$ irrational $\rightarrow$ aperiodic (**67 Mng**).

Show that a discrete-time signal is periodic in frequency with period 2π 2 4

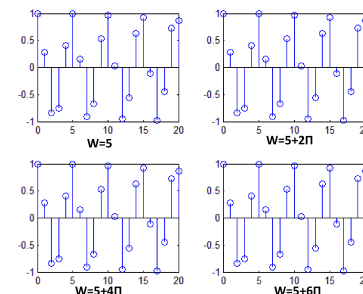
PIN 2/45 69 Bh, 80 Bh · also the reason every DTFT and DFT plot is drawn over one 2π span

- Raise the frequency by 2π :

$$e^{j(\omega_0+2\pi)n} = e^{j2\pi n} e^{j\omega_0 n} = e^{j\omega_0 n}$$

because $e^{j2\pi n} = 1$ for integer n .

- So $\omega_0, \omega_0 \pm 2\pi, \omega_0 \pm 4\pi \dots$ are the **same sequence**, sample for sample.
- Consequences worth one line each:
 - All distinct DT frequencies lie in one span of 2π , usually $-\pi \leq \omega < \pi$.
 - $\omega = \pi$ is the **highest** rate of oscillation (alternating $+1, -1$); $\omega = 0$ and $\omega = 2\pi$ are both DC.
 - In continuous time nothing like this happens: $e^{j\Omega t}$ is distinct for every Ω .



$\cos(\omega n)$ at $\omega = 5, 5 + 2\pi, 5 + 4\pi, 5 + 6\pi$: four different formulas, one sequence. BB Sir, deck 1.1

1.5 Transformation of the independent variable

Plot the transformed sequence, Eg $x[-2n + 3]$ or $u[n] - u[n - 3] + 5\delta[n - 4] + nu[n - 6]$ 2 3 3+2

HOT 4/45 76 Ch, 74 Ch, 66 Ma, 69 Bh · pure plotting marks · every instance is worked in Ch1-num §8

The three operations

- **Shift** $x[n - k]$: delay by k (move right) for $k > 0$; advance for $k < 0$.
- **Fold (time reversal)** $x[-n]$: mirror about $n = 0$.
- **Scale** $x[an]$:
 - $|a| > 1$ integer = **decimation**, keep every a -th sample, **samples are lost**.
 - $|a| < 1$ = interpolation, new samples must be filled in.
 - **Not reversible like the CT case**: this is why down-sampling needs its own treatment.
- **Amplitude operations**: addition, multiplication, scaling act sample by sample on aligned indices.

Order of operations for $x[an + b]$ (where marks are lost)

1. **Shift first, then scale/fold**, or you will shift by the wrong amount.
2. Safer still: work index by index. Set $m = an + b$, then for each sample $x[m]$ of the original, its new position is $n = (m - b)/a$.
3. Keep only integer n ; a non-integer means that sample disappears.
4. **Check one point**: put $n = 0$ into $an + b$ and confirm the value lands where you drew it.

Worked index map: $x[-2n + 3]$, $x[n] = \{1, 2, 0, -1, -3, -4\}$ starting at $n = 0$

- $n = 0 \rightarrow x[3] = -1$; $n = 1 \rightarrow x[1] = 2$;
 $n = -1 \rightarrow x[5] = -4$; $n = 2 \rightarrow x[-1] = 0$.
- Only odd-indexed original samples survive: folding + decimation by 2.

1.6 Discrete-time Fourier series (DTFS)

Explain how the Fourier series coefficients are calculated 3 4

PIN 2/45 73 Shr, 72 Ch · 76 Ash adds Dirichlet's conditions, see §1.7

The pair

- A periodic $x[n]$ of period N has fundamental frequency $\omega_0 = 2\pi/N$.

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n} \quad (\text{synthesis})$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n} \quad (\text{analysis})$$
- $\langle N \rangle$ means *any* N consecutive values, usually 0 to $N - 1$.
- $a_k = a_{k+N}$: only N distinct coefficients exist, because $e^{j(k+N)\omega_0 n} = e^{jk\omega_0 n}$.
- That is the one real difference from the continuous-time Fourier series, which has infinitely many distinct coefficients.

Deriving a_k , if the question says “explain the process”

1. Start from the synthesis equation.
2. Multiply both sides by $e^{-jr(2\pi/N)n}$ and sum over one period.
3. Interchange the two summations.
4. Use the orthogonality result

$$\sum_{n=\langle N \rangle} e^{j(k-r)(2\pi/N)n} = \begin{cases} N, & k - r = 0, \pm N, \pm 2N \dots \\ 0, & \text{otherwise} \end{cases}$$

5. Every term dies except $k = r$, leaving $\sum x[n] e^{-jr(2\pi/N)n} = a_r N$.

Method for a numerical (faster than the integral)

1. Find N first (§1.4), then $\omega_0 = 2\pi/N$.
2. Expand every cos/sin with Euler's formula.
3. Force each exponent into the form $jk(2\pi/N)n$ and read a_k off by inspection.
4. Use $a_k = a_{k+N}$ for anything outside $0 \leq k \leq N - 1$.

Properties of the DTFS ($x[n]$, $y[n]$ both period N)

Property	Signal	Coefficient
Linearity	$Ax[n] + By[n]$	$Aa_k + Bb_k$
Time shift	$x[n - n_0]$	$a_k e^{-jk(2\pi/N)n_0}$
Frequency shift	$e^{jM(2\pi/N)n}x[n]$	a_{k-M}
Time reversal	$x[-n]$	a_{-k}
Conjugation	$x^*[n]$	a_k^*
Periodic convolution	$\sum_{r=\langle N \rangle} x[r]y[n-r]$	$N a_k b_k$
Multiplication	$x[n]y[n]$	$\sum_{l=\langle N \rangle} a_l b_{k-l}$
Parseval	$\frac{1}{N} \sum_{n=\langle N \rangle} x[n] ^2$	$\sum_{k=\langle N \rangle} a_k ^2$

Conjugate symmetry, asked as a rider: if $x[n]$ is real then $a_{-k} = a_k^*$. Real and even $\rightarrow a_k$ real and even; real and odd $\rightarrow a_k$ imaginary and odd with $a_0 = 0$. Parseval reads: total average power in one period = sum of the average powers $|a_k|^2$ of all harmonics.

1.7 Discrete-time Fourier transform (DTFT)

Explain the Fourier transform multiplication property for two sequences; write Dirichlet's conditions

4+3

PIN 1/45 76 Ash · the DTFT itself is assumed knowledge in every Ch 3 question, so this section is a prerequisite more than a target

The pair

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}, \quad x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})e^{j\omega n} d\omega$$

- Derived from the DTFS by letting $N \rightarrow \infty$: the discrete lines Na_k merge into the continuous envelope $X(e^{j\omega})$.
- $X(e^{j\omega})$ is periodic in ω with period 2π , because $e^{-jn(\omega+2\pi)} = e^{-j\omega n}$. So the inverse integral can be taken over any 2π span.
- Exists when $x[n]$ is absolutely summable, $\sum |x[n]| < \infty$.

Dirichlet's conditions (quote all three)

- $x[n]$ is absolutely summable, $\sum_n |x[n]| < \infty$.
- $x[n]$ has a finite number of maxima and minima in any finite interval.
- $x[n]$ has a finite number of finite discontinuities in any finite interval.

Sufficient, not necessary: $u[n]$ fails condition 1 yet has a DTFT once impulses in ω are allowed.

Properties of the DTFT

Property	Signal	Transform
Linearity	$ax_1[n] + bx_2[n]$	$aX_1(e^{j\omega}) + bX_2(e^{j\omega})$
Time shift	$x[n - n_0]$	$e^{-j\omega n_0} X(e^{j\omega})$
Frequency shift	$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega-\omega_0)})$
Time reversal	$x[-n]$	$X(e^{-j\omega})$
Conjugation	$x^*[n]$	$X^*(e^{-j\omega})$
Convolution	$x[n] * h[n]$	$X(e^{j\omega})H(e^{j\omega})$
Multiplication	$x_1[n]x_2[n]$	periodic convolution of the spectra, $\div 2\pi$
Parseval	$\sum_n x[n] ^2$	$\frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) ^2 d\omega$

If $x[n]$ is real, $X(e^{j\omega})$ is conjugate symmetric: magnitude even, phase odd. Real and even $\rightarrow$ transform real and even; real and odd $\rightarrow$ transform imaginary and odd. **Parseval here says energy, not power**: it is the DTFS version that talks about power.

Multiplication property, and why it is not a plain product

- Convolution in one domain = multiplication in the other, both ways:

$$x[n] * h[n] \longleftrightarrow X(e^{j\omega})H(e^{j\omega})$$

$$x_1[n]x_2[n] \longleftrightarrow \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\theta})X_2(e^{j(\omega-\theta)}) d\theta$$

- The second is a **periodic (circular) convolution** of the two spectra, scaled by $1/2\pi$, integrated over any 2π interval.
- It has to be periodic**: both spectra are 2π -periodic, so an ordinary infinite convolution would diverge.
- Proof in one move: substitute the inverse DTFT of $x_1[n]$ into the forward DTFT of the product, then swap the sum and the integral.
- This is the windowing property used to explain Gibbs oscillation in Ch 5: truncating $h[n]$ with a window multiplies in time, so it convolves in frequency and smears the band edge.

1.8 Properties of a discrete-time system

Test the given system for linearity, time-invariance, causality, memory and BIBO stability **3 5 10**

TOP 12/45 76 Ch, 74 Ch, 82 Ba, 78 Ch, 75 Ash, 71 Bh, 70 Bh, 68 Bh, 67 Mng, 66 Ma, 69 Bh, 80 Bh · plus 81 Ba on BIBO alone · every instance worked in Ch1-num §9

The six properties, with the test that proves each

- **Static / memoryless:** $y[n]$ depends only on $x[n]$ at the same n .
 - Test: does any shifted index $x[n - k]$, $x[2n]$ or past output appear? If yes, **dynamic**.
- **Invertible:** distinct inputs give distinct outputs, so an inverse system exists with $h[n] * h_1[n] = \delta[n]$.
- **Causal:** $y[n]$ uses only present and past inputs.
 - Test: look for $x[n + k]$, $k > 0$, or a folded index. $y[n] = x[-n]$ is **non-causal**: at $n = -2$ it needs $x[2]$.
- **Time invariant:** shift the input, the output shifts identically.
- **Linear:** superposition holds.
- **BIBO stable:** every bounded input gives a bounded output.

BIBO stability

- Bounded means $|x[n]| \leq M_x < \infty$ for all n .
- **To disprove:** find one bounded input giving an unbounded output. A constant or $u[n]$ almost always works.
 - Eg $y[n] = nx[n]$ with $x[n] = 1$ gives $y[n] = n \rightarrow \infty \times$ unstable.
 - Eg $y[n] = \sum_{k=0}^n x[k]$ (accumulator) with $x[n] = u[n]$ gives $y[n] = n + 1 \times$ unstable. That is **80 Bh** in one line.
- **To prove** for an LTI system, use the impulse-response test in §1.9 instead: $\sum_k |h[k]| < \infty$.
- Recursive example, **81 Ba**: $y[n] = x[n] + e^a y[n - 1]$ has $h[n] = (e^a)^n u[n]$, so it is stable only if $|e^a| < 1$, i.e. **$a < 0$** .

Linearity, the only reliable way to write it

1. $x_1[n] \rightarrow y_1[n]$ and $x_2[n] \rightarrow y_2[n]$.
 2. Apply $x[n] = a x_1[n] + b x_2[n]$ and compute $y[n]$ from the system equation.
 3. Compare against $a y_1[n] + b y_2[n]$. Equal $\rightarrow$ linear.
- Eg $y[n] = nx[n]$: $n(ax_1 + bx_2) = a n x_1 + b n x_2 \checkmark$ linear. $y[n] = x^2[n]$: the cross term $2abx_1x_2$ survives $\times$ not linear.

Time invariance, the only reliable way to write it

1. Delayed input: put $x_1[n] = x[n - n_0]$ into the equation, call the result $y_1[n]$.
2. Delayed output: write $y[n - n_0]$ by replacing every n in $y[n]$, **including the explicit ones**.
3. Equal $\rightarrow$ time invariant.

Eg $y[n] = nx[n]$: $y_1[n] = nx[n - n_0]$ but $y[n - n_0] = (n - n_0)x[n - n_0] \times$ time varying. $y[n] = \sin(x[n]) \checkmark$ time invariant. **An explicit n multiplying the input is the classic time-varying flag**; folding $y[n] = x[-n]$ is time varying too.

Answer skeleton **5 10**

1. State the property in one line.
2. Apply its test to *this* system, showing the algebra.
3. Give the verdict with the reason attached, Eg “not time invariant, because the explicit n is not delayed”.

Marks are awarded per property, so answer all the parts asked even if one is obvious. A bare “yes/no” with no working scores nothing.

Results worth memorising

- $y[n] = x[-n]$: linear $\checkmark$, time varying $\times$, non-causal $\times$, stable $\checkmark$.
- $y[n] = x[n^2]$ or $x[2n]$: linear $\checkmark$, time varying $\times$.
- $y[n] = x^2[n]$, $e^{x[n]}$, $\log x[n]$: **non-linear** $\times$, but time invariant $\checkmark$ and memoryless $\checkmark$.
- $y[n] = y[n - 4] + x[n - 4]$: linear $\checkmark$, time invariant $\checkmark$, causal $\checkmark$, **dynamic and unstable**.
- $y[n] = \cos(\frac{5\pi}{8}n + \frac{\pi}{4})$ has no input at all: not a system response, so it fails linearity trivially.

1.9 LTI systems and the convolution sum

What is the convolution summation? Derive its equation and explain its significance 1+3 2 3

HOT 4/45 69 Bh, 73 Ma, 70 Asa, 70 Bh · the theory half of the subject's most-asked question

Derivation, four steps (**this is the full-mark answer**)

1. Decompose the input into impulses. By the sifting property,

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

i.e. $\dots + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + \dots$

2. Use time invariance. If $\delta[n] \rightarrow h[n]$, then $\delta[n-k] \rightarrow h[n-k]$.

3. Use linearity. The input is a weighted sum of shifted impulses, so the output is the same weighted sum of shifted impulse responses.

4. Therefore

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] = x[n] * h[n]$$

Significance, if asked 2

- $h[n]$ **alone characterises an LTI system completely**. Know the impulse response, know the output for every possible input.
- It converts a system description into an arithmetic operation, so the response can be computed rather than solved.
- Causality and stability both become simple tests on $h[n]$ (below).

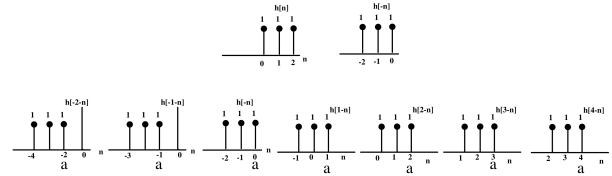
Properties of LTI systems

- **Identity**: $x[n] * \delta[n] = x[n]$.
- **Shifting**: $x[n] * \delta[n-k] = x[n-k]$.
- **Commutative**: $x[n] * h[n] = h[n] * x[n]$: fold whichever sequence is shorter.
- **Distributive**: $x * (h_1 + h_2) = x * h_1 + x * h_2 \rightarrow$ two systems in **parallel** add their impulse responses.
- **Associative**: $x * (h_1 * h_2) = (x * h_1) * h_2 \rightarrow$ two systems in **cascade** convolve their impulse responses, and their order does not matter.

Proof of the stability condition, if asked to derive it

- $|y[n]| = |\sum_k h[k]x[n-k]| \leq \sum_k |h[k]| |x[n-k]|$ by the triangle inequality and $|ab| = |a||b|$.
- Bound the input, $|x[n-k]| \leq M_x$: $|y[n]| \leq M_x \sum_k |h[k]|$.
- So the output is bounded whenever $\sum_k |h[k]| < \infty$.

Worked cases: $h[n] = \delta[n - n_0]$ gives $\sum |h| = 1$, stable. $h[n] = u[n - n_0]$ gives $\sum |h| = \infty$, unstable. $h[n] = a^n u[n]$ gives $\sum |h| = 1/(1 - |a|)$, **stable only if $|a| < 1$** .



Building $h[a-n]$: fold $h[n]$, then slide. Each value of a gives one output sample. BB Sir, deck 3

The graphical method, four steps

1. Fold $h[k] \rightarrow h[-k]$.
2. Shift by n : $h[n-k]$.
3. Multiply with $x[k]$ sample by sample.
4. Sum the products. That single number is $y[n]$. Repeat for each n .

Shortcut for two finite sequences: the output starts at $n_x + n_h$ (sum of the starting indices) and has $L_x + L_h - 1$ samples. Use it to check your answer's length and position, every time.

- **Memoryless** $\Leftrightarrow h[n] = k \delta[n]$, giving $y[n] = kx[n]$.
- **Invertible** $\Leftrightarrow$ there is $h_1[n]$ with $h[n] * h_1[n] = \delta[n]$.
- **Causal** $\Leftrightarrow h[n] = 0$ for $n < 0$.
 - From $y[n] = \sum_k h[k]x[n-k]$: terms with $k < 0$ would need future inputs.
- **BIBO stable** $\Leftrightarrow \sum_k |h[k]| < \infty$ (absolutely summable).
- **Step response** $s[n] = h[n] * u[n] = \sum_{k=-\infty}^n h[k]$, so $h[n] = s[n] - s[n-1]$.

Differentiate FIR and IIR systems; define recursive and non-recursive systems with examples 3+3 4

5

HOT 4/45 74 Bh, 70 Ma, 72 Ash, 80 Ba · the FIR/IIR comparison is re-used in Ch 3 and Ch 4, learn it once

FIR Vs IIR

	FIR	IIR
$h[n]$	finite, $h[n] = 0$ outside $0 \leq n \leq M - 1$	infinite length
Output Memory	$\sum_{k=0}^{M-1} h[k]x[n-k]$ finite, last M samples	$\sum_{k=0}^{\infty} h[k]x[n-k]$ infinite
Built with	convolution directly	difference equation, feedback
Stability	always stable	only if poles are inside the unit circle
Phase	can be exactly linear	cannot be exactly linear

The reason IIR needs a difference equation: convolution would need infinite memory, so it cannot be implemented directly.

Recursive Vs non-recursive

- **Non-recursive:** output from present and past *inputs* only, $y[n] = F\{x[n], \dots, x[n-M]\}$. **No feedback.** A causal FIR system is non-recursive.
 - Eg moving average
 $y[n] = \frac{1}{3}\{x[n] + x[n-1] + x[n-2]\}$.
- **Recursive:** output also uses past *outputs*, $y[n] = F\{y[n-1], \dots, y[n-N], x[n], \dots, x[n-M]\}$. **Has a feedback loop.**
 - Eg $y[n] = x[n] + a y[n-1]$.
 - Recursive form of the moving average (asked in 70 Ma): $y[n] = y[n-1] + \frac{1}{M}\{x[n] - x[n-M]\}$: one add and one subtract per sample instead of M adds.
- **FIR/IIR is about the impulse response; recursive/non-recursive is about the implementation.** They usually coincide but are not the same statement: a recursive structure can still realise an FIR response.

State the linear constant-coefficient difference equation; find the zero-input and zero-state response 4 5

HOT 5/45 80 Ba, 76 Bh, 77 Ch, 73 Ch, 81 Ba · the system-function half is in Ch 3, the response half is worked in Ch1-num §11

General form

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k], \quad a_0 = 1$$

$$\text{or } y[n] = -\sum_{k=1}^N a_k y[n-k] + \sum_{k=0}^M b_k x[n-k]$$

- N is the **order** of the system.
- $N = 0 \rightarrow$ non-recursive (FIR); $N \geq 1 \rightarrow$ recursive.

Total response = zero-input + zero-state

- Iterating $y[n] = a y[n-1] + x[n]$ from $y[-1]$ gives

$$y[n] = \underbrace{a^{n+1}y[-1]}_{\text{zero input}} + \underbrace{\sum_{k=0}^n a^k x[n-k]}_{\text{zero state}}$$

- **Zero-state (forced) response:** set $y[-1] = 0$. The part the input forces on the system. A system with zero initial conditions is called **relaxed**.
- **Zero-input (natural, free) response:** set $x[n] = 0$. Depends only on the system and its initial conditions.
- **An LTI system must be relaxed.** With non-zero initial conditions the system is not linear, which is why questions say “initially relaxed”.

Find the frequency response of an LTI system / its response to a complex exponential 4 5

HOT 7/45 70 Ch, 72 Ka, 80 Ba, 71 Shr, 69 Bh, 73 Shr, 82 Ba · the cheapest 5 marks in the paper, see the shortcut · worked in Ch1-num §3

- Feed $x[n] = e^{j\omega n}$ into the convolution sum:

$$y[n] = \sum_k h[k] e^{j\omega(n-k)} = e^{j\omega n} \underbrace{\sum_k h[k] e^{-j\omega k}}_{H(e^{j\omega})}$$

- So $y[n] = H(e^{j\omega}) e^{j\omega n}$: **the input comes out unchanged in shape, only scaled.**
- $e^{j\omega n}$ is therefore an **eigenfunction** of every LTI system, and $H(e^{j\omega})$, the DTFT of $h[n]$, is the **eigenvalue**, called the **frequency response**.
- $|H(e^{j\omega})|$ is the magnitude response, $\angle H(e^{j\omega})$ the phase response. Both are 2π -periodic (§1.4).

The shortcut (do not convolve these)

- The question gives $x[n] = Ae^{j\omega_0 n}$ over all n , so it is **not a finite sequence**: graphical convolution is the wrong tool.
- Compute $H(e^{j\omega})$ once. For $h[n] = a^n u[n]$:

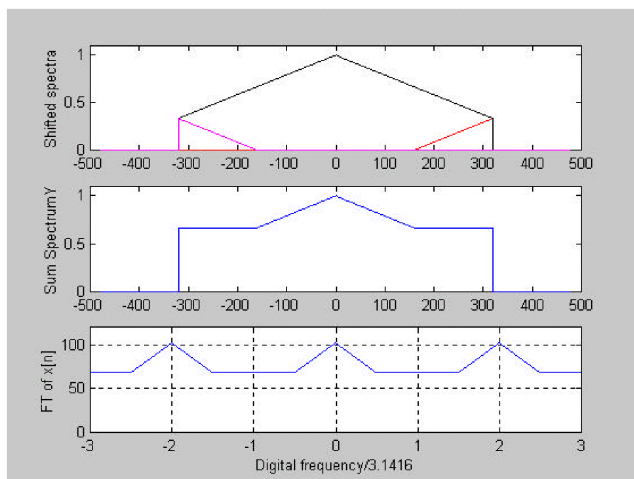
$$H(e^{j\omega}) = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \frac{1}{1 - ae^{-j\omega}}$$

- Evaluate at the ω_0 the question gives.
- Answer: $y[n] = A H(e^{j\omega_0}) e^{j\omega_0 n}$.

For a sum of sinusoids (73 Shr) do this once per frequency and add: superposition. For a real cosine, write it as two exponentials at $\pm\omega_0$ and use conjugate symmetry.

1.10 Sampling of continuous-time signals

What is sampling? How is the spectrum of the sampled signal related to that of the continuous-time signal? Explain aliasing 6 8

PIN 2/45 69 Bh, 68 Bh · the numerical form, "write $x[n]$ after sampling", is asked by 7 more papers, worked in Ch1-num §10

Shifted copies, their sum, and the resulting $X(e^{j\omega})$. The copies overlap here, so the triangle is distorted: **this is aliasing**. Chapterwise Ch2 p43

The relation (quote this equation)

- Sampling means $x[n] = x_c(nT)$, with T the sampling period and $F_s = 1/T$ the sampling rate.
- Substituting into the inverse CTFT and regrouping gives

$$X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\frac{\omega}{T} + j\frac{2\pi k}{T}\right)$$

- In words: **the spectrum of the sampled signal is the spectrum of the analog signal, scaled by $1/T$, repeated every $\Omega_s = 2\pi/T$** , with the analog frequency mapped to $\omega = \Omega T$.
- Three-step recipe for the diagram: (1) cut $X_c(j\Omega)$ into bands of width $2\pi/T$; (2) shift every band to the centre; (3) add.

Aliasing

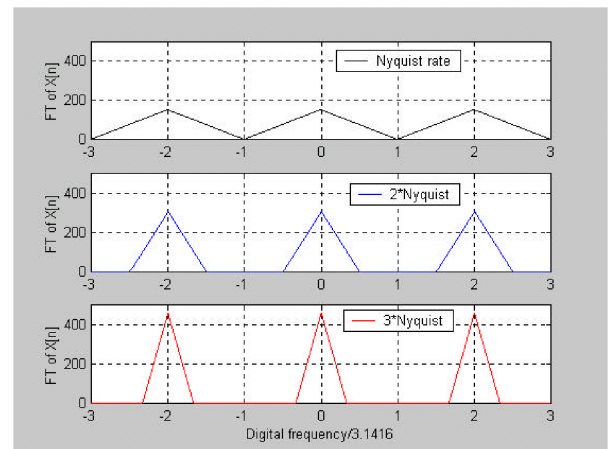
- If the copies are spaced closer than the signal is wide, they **overlap** and add. The overlapped region can no longer be separated.
- A high frequency then reappears as, and is indistinguishable from, a lower one.
- Cure**: sample faster, or band-limit the input first.

Sampling theorem

- Aliasing disappears when

$$\Omega_s = \frac{2\pi}{T} \geq 2W \quad \text{or} \quad F_s \geq 2F_{max}$$

- $2F_{max}$ is the **Nyquist rate**; $F_s/2$ is the **folding frequency**.
- **Oversampling** means F_s is taken well above the Nyquist rate: it pushes the copies further apart, so the anti-aliasing filter can be gentler (76 Bh asks exactly this).
- When the criterion holds, every copy but $k = 0$ is zero in $-\pi \leq \omega < \pi$, so $X(e^{j\omega}) = \frac{1}{T} X_c(j\omega/T)$: **the shape is preserved**, only scaled. That is why perfect reconstruction is possible.
- The relative width of the digital spectrum is $\rho = 2W/\Omega_s$: the faster you sample, the narrower the signal sits inside the 2π span.



Same signal at the Nyquist rate, twice it, three times it. The copies pull apart and the gap between them grows. Chapterwise Ch2 p45

Why is an anti-aliasing filter needed? Describe the effect of the sample-and-hold circuit at the input of an A/D converter 2+5

PIN 1/45 69 Bh · EX 753 only, but it is the theory behind every sampling numerical

Anti-aliasing filter

- A real signal is never perfectly band-limited: noise and harmonics run above $F_s/2$.
- Those components would fold down into the band of interest and **cannot be removed afterwards**: once aliased, the damage is permanent.
- So an **analog low pass filter** is placed *before* the sampler, with cut-off at or below $F_s/2$.
- It is analog by necessity: a digital filter would come after the sampling that caused the problem.
- Practical filters are not brickwall, which is the second reason to oversample: it widens the transition band the filter is allowed to use.

Sample-and-hold

- An A/D converter needs a **constant** input during its conversion time. A signal still moving would be converted to a wrong, smeared value.
- The S/H circuit tracks the input, then freezes it at the sampling instant and holds it flat until the conversion finishes.
- Effects to state:
 - Holding is a zero-order hold: it multiplies the spectrum by a sinc-shaped envelope, so high frequencies droop.
 - That droop is corrected later by a digital $\times 1/\text{sinc}$ equaliser.
 - Real S/H adds aperture jitter, droop during hold and a small pedestal error.
- **Reconstruction** at the output is the mirror image: D/A holds each sample flat, then an analog *smoothing* low pass filter removes the spectral images.

1.11 Last-minute recall for this chapter

Must memorise

- $x_e = \frac{1}{2}(x[n] + x[-n])$, $x_o = \frac{1}{2}(x[n] - x[-n])$.
- Periodic $\Leftrightarrow \omega_0/2\pi$ rational; then $N = 2\pi m/\omega_0$. **No π in ω_0 means aperiodic.**
- Energy signal: $0 < E < \infty$, $P = 0$. Power signal: $0 < P < \infty$, $E = \infty$. Periodic $\rightarrow$ power. $|e^{j\theta}| = 1 \rightarrow$ power, $P = 1$.
- E for $a^n u[n]$ is $1/(1 - a^2)$; P for $u[n]$ is $\frac{1}{2}$.
- $y[n] = \sum_k x[k]h[n - k]$. Output starts at $n_x + n_h$, length $L_x + L_h - 1$.
- LTI causal $\Leftrightarrow h[n] = 0$, $n < 0$. LTI stable $\Leftrightarrow \sum |h[k]| < \infty$.
- $e^{j\omega n}$ in $\rightarrow H(e^{j\omega})e^{j\omega n}$ out. Never convolve these.
- $X(e^{j\omega}) = \frac{1}{T} \sum_k X_c(j\frac{\omega + 2\pi k}{T})$, and $F_s \geq 2F_{max}$.

Most repeated, in order

1. Convolution, find $y[n]$ **TOP 34/45**
2. Energy Vs power + periodicity **TOP 10/45**
3. Test a system for linearity / TI / causality / stability **TOP 12/45**
4. Find the fundamental period **HOT 7/45**
5. Response to a complex exponential **HOT 7/45**
6. Sample an analog signal, write $x[n]$ **HOT 7/45**
7. Convolution from tabulated or stem-plot data **HOT 4/45**
8. Even and odd parts **HOT 4/45**
9. FIR Vs IIR, recursive Vs non-recursive **HOT 4/45**
10. Convolution sum, derive and explain **HOT 4/45**

Q1 and Q2 of the paper come from this chapter in almost every year. If time is short, do §1.9 and §1.3 and leave §1.6: the DTFS has been asked 5 times in 45 papers and never for more than 6 marks.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. $\rightarrow$ reads “leads to / therefore”, $+$ reads “and”, $=$ reads “is”. $\checkmark$ means the property holds, $\times$ that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months:** 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II).
Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 Bh is EX 753 Regular, 74 Ma is EX 753 Back.
- **69 Bh and 69 Bh are two different papers**, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.